Jingwen Zhang

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BACKEND & AI INFRASTRUCTURE ENGINEER

Software Engineer and former Technical Lead with 3+ years of experience building production systems across backend, Kubernetes, IoT, and embedded environments. Currently completing an M.S. in Data Science at UC San Diego, with a focus on ML systems, cloud platforms, and production-oriented AI infrastructure.

EDUCATION

University of California San Diego — San
M.S. in Data Science — Sep 2025 – Dec 2026
Selected Coursework: Machine Learning Systems, Deep Learning Systems, Scalable Database Systems

University of Utah — Salt Lake City, UT
B.S. in Computer Science — May 2019 – Dec 2022

PROFESSIONAL EXPERIENCE

iLumens LLC — Salt Lake City, UT
Technical Lead — May 2024 – Jun 2025
Software Engineer — Dec 2022 – May 2024

- Promoted to **Technical Lead** for an 8-engineer product team; owned architecture decisions, technical decomposition, code reviews, release planning, and cross-functional delivery across backend, frontend, cloud, and firmware workstreams.
- Led a **full-stack performance overhaul** across embedded firmware, Django backend, GKE-hosted services, and React frontend; personally identified and resolved bottlenecks at each layer, delivering a **system-wide speed improvement exceeding 5x**.
- Architected and operated a cloud-based **IoT telemetry platform** on **GCP** using Django, PostgreSQL, Docker, and **Kubernetes (GKE)**; built **CI/CD pipelines**, Cloud SQL connectivity, Kubernetes Ingress/TLS, IAM, and multi-zone deployments.
- Led **migration from Astarte to ThingsBoard**, coordinating backend refactoring, telemetry schema alignment, and phased rollout across production environments.
- Architected and implemented a **C#/.NET** real-time training application for power-grid operators under a **U.S. Department of Energy** funded program; integrated a live simulation engine through custom DLL interop and coordinated requirements across simulation engineers and cognitive scientists.
- Developed **embedded firmware in C** for ESP32 and nRF microcontrollers, handling hardware-level data acquisition in memory constrained environments and integrating device telemetry with cloud services.

Intermountain Healthcare — Salt Lake City, UT
Information Systems Apprentice — May 2022 – Aug 2022

- Developed internal web applications using **C#**, .NET Core, Vue.js, and SQL to support operational workflows across an organization of approximately **59,000 employees**.

SELECTED PROJECTS

MiniServe — LLM Serving Scheduler — Jun 2026 – Present

- Designed a **Python-based LLM serving simulator** with request admission, prefill/decode scheduling, state transitions, and capacity-aware multi-request execution.

High-Performance LLM Inference Benchmarking — Jan 2026 – Apr 2026

- Benchmarked long-context **Qwen2-7B** inference with **vLLM**, SDPA, and **SnapKV** on NVIDIA A100 GPUs, analyzing throughput, latency, and KV-cache memory trade-offs.

GPU Kernel Optimization & Distributed Transformer Parallelism — Sep 2025 – Dec 2025

- Implemented a fused **Triton** matmul-add-ReLU kernel and **MPI**-based tensor-parallel communication routines, including custom collectives, all-gather, all-reduce, and reduce-scatter.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, TypeScript/JavaScript, SQL, Java, Bash

Backend & Applications: Django, .NET Core, PostgreSQL, Cloud SQL, REST APIs, React

Cloud & Platform: GCP, GKE, Kubernetes, Docker, IAM, CI/CD, TLS, Kubernetes Ingress, Linux

ML Systems: PyTorch, vLLM, Triton, SDPA, KV-cache optimization, AWQ, INT4/INT8 quantization

Embedded & Systems: Embedded C, ESP32, nRF, device-to-cloud integration